

ECE 3337: Signals & Systems Analysis

Lecture 25: Fourier Applications

**Reference: Chapter 6
(Section 6.4, 6.5, 6-8)**



Final Exam

Date: May 11th 2 - 4pm, SW 102 (Social Work Building).

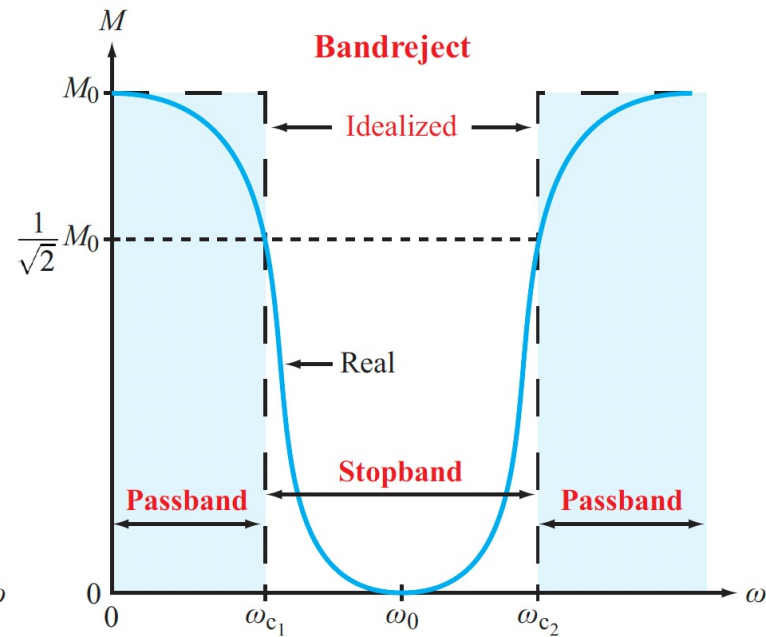
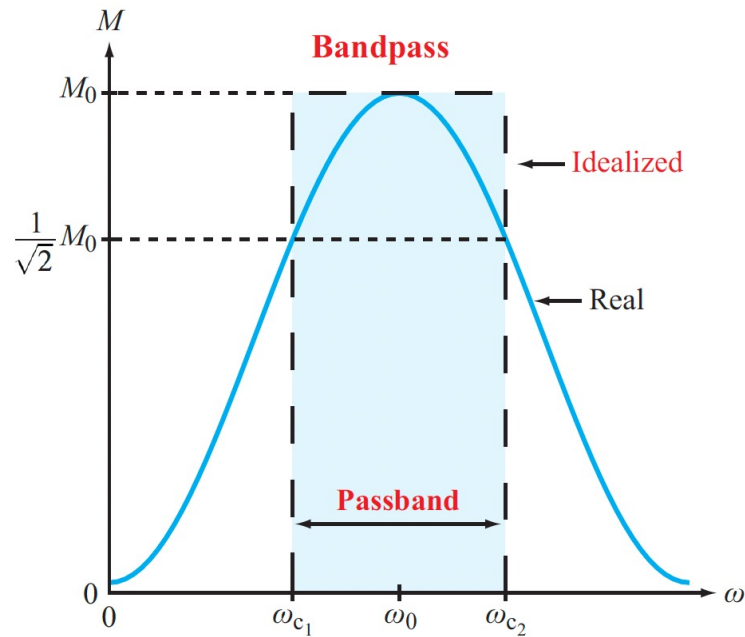
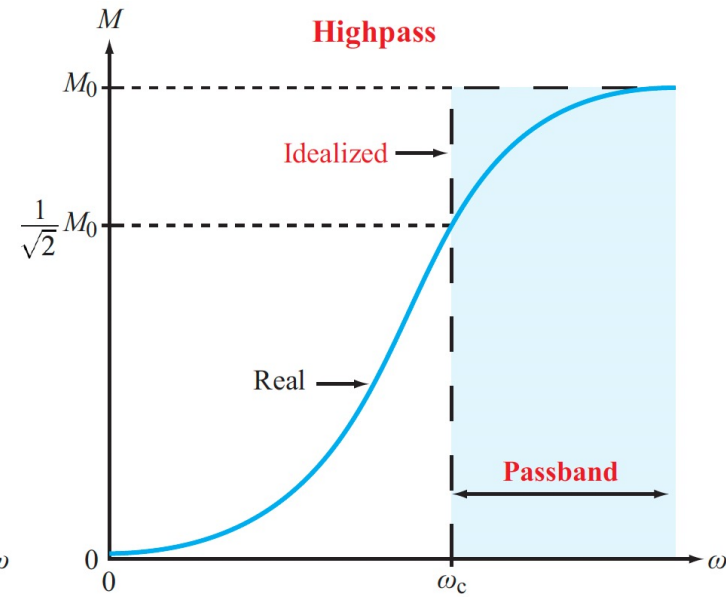
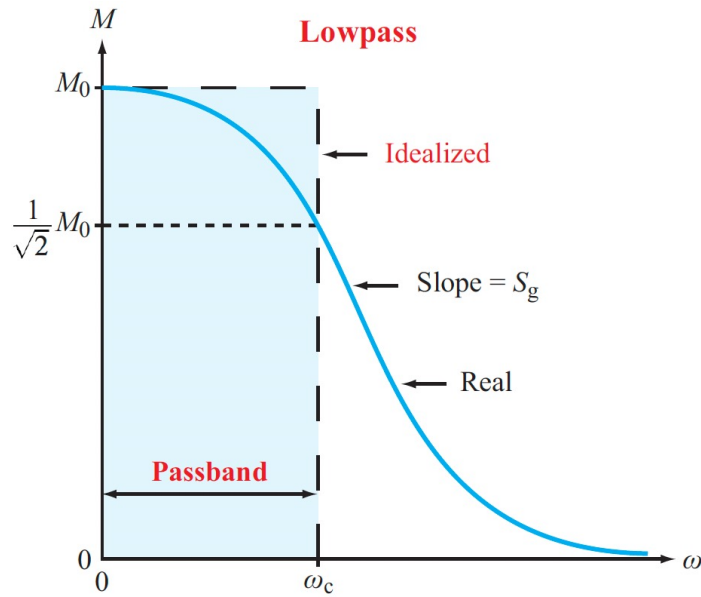
<https://www.uh.edu/infotech/services/facilities-equipment/supported-classrooms/sw/102/>

Plenty of old exams are posted on MS Teams for practice

- Practice for accuracy and speed
- Test yourself individually, rather than as a group
- Avoid a false sense of knowing a topic
 - This can easily happen in a group setting
- Suggested study strategy:
 - Do the practice tests individually
 - Identify the challenging problems, and ask your instructors for help
 - Discuss the challenging problems in your study group
 - Follow up with individual work to confirm your understanding



Recap: Common Basic Filters

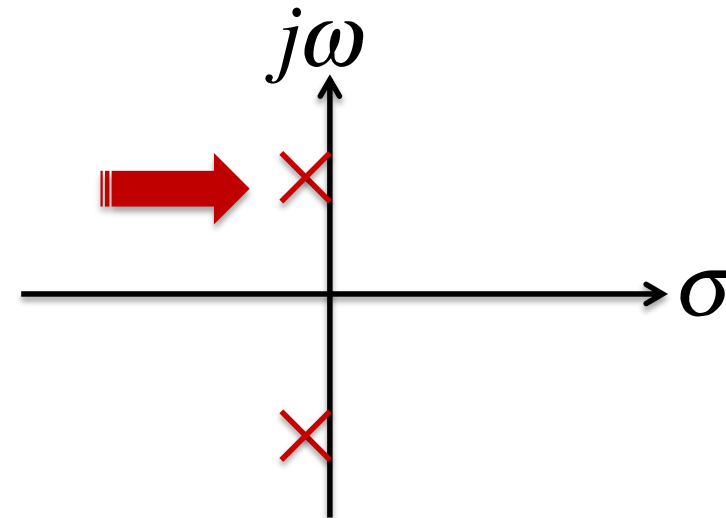
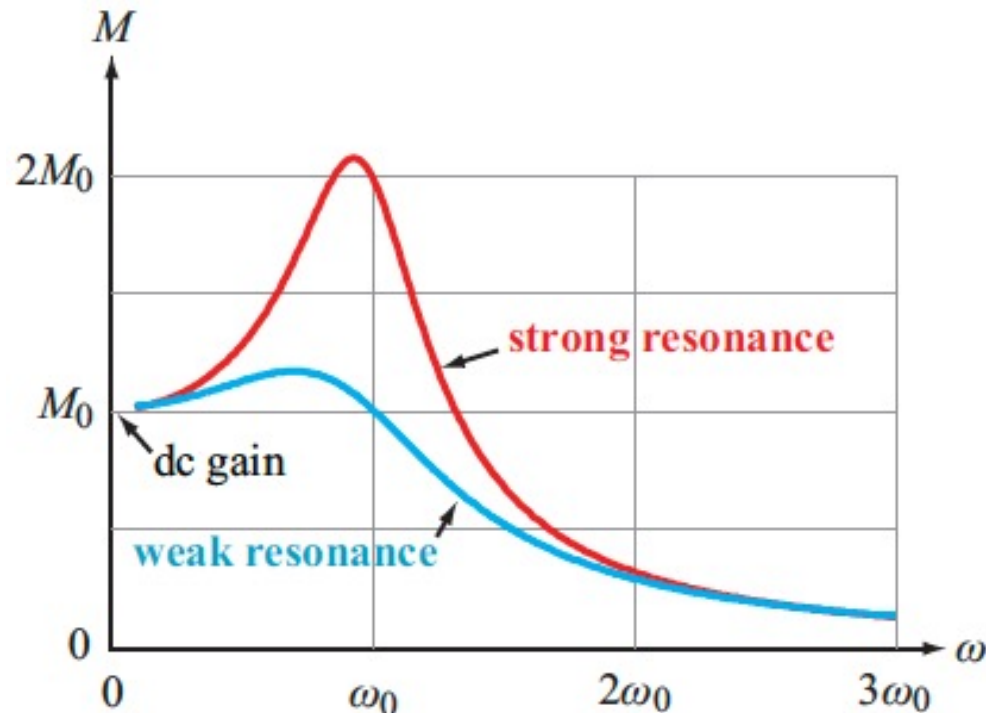




Recap: Resonance

Resonance happens when 3 conditions are satisfied:

1. The circuit / system has an oscillatory mode (a pair of complex conjugate poles) with a natural frequency ω_0
2. The applied input is at the natural frequency ω_0 (supplies energy)
3. The resistance (damping) is negligible, i.e., the system is “barely stable”, and the poles are close to, or on the vertical axis





Bode Plots for Filters

Hendrik Bode showed that sketching the frequency response of systems (magnitude and phase) becomes super convenient if we adopt logarithmic scales

- We can do them easily by hand, even for complicated systems, by using Bode's method
- Usually, our systems are expressed as a ratio of factored polynomials, for example:

$$\hat{\mathbf{H}}(j\omega) = \frac{K(j\omega + a_1)}{j\omega(j\omega + b_1)((j\omega)^2 + b_2j\omega + b_3)}$$



Bode's Awesome Idea

$$\hat{\mathbf{H}}(j\omega) = \left(\frac{Ka_1}{b_1b_3} \right) \frac{\left(\frac{j\omega}{a_1} + 1 \right)}{(j\omega) \left(\frac{j\omega}{b_1} + 1 \right) \left(\frac{(j\omega)^2}{b_3} + \frac{j\omega}{(b_3/b_2)} + 1 \right)}$$

**NORMALIZED
FORMULA**

$$\begin{aligned} \Rightarrow \log |\hat{\mathbf{H}}(j\omega)| &= \log \left| \frac{Ka_1}{b_1b_3} \right| + \log \left| \frac{j\omega}{a_1} + 1 \right| \\ &\quad - \log |j\omega| - \log \left| \frac{j\omega}{b_1} + 1 \right| - \log \left| \frac{(j\omega)^2}{b_3} + \frac{j\omega}{(b_3/b_2)} + 1 \right| \end{aligned}$$

**IF WE LEARN TO PLOT EACH OF THESE BASIC TYPES OF TERMS,
WE CAN PLOT THE FULL RESPONSE BY ADDITION AND SUBTRACTION!**



The Logarithmic Scale

The human basis:

- The human senses (sight, hearing, touch, ...) respond in a logarithmic manner
- The human senses respond to ratios of signal power, not increments
 - We perceive a sound S_2 as being twice as loud as sound S_1 if the ratio of their powers is 10

$$\log_{10} \left(\frac{P_2}{P_1} \right) = 1 \quad \text{when } P_2 = 10P_1$$

- **Bode wisely made two choices:**
 - He chose to use base 10 logarithms
 - He adopted the *decibel* unit, used for ratios



The Decibel: Log Power Ratio

Given two signals S_1 and S_2 , whose powers P_1 and P_2 are such that: $P_2 = 10P_1$, the log ratio is given by:

$$\log_{10} \left(\frac{P_2}{P_1} \right) = 1 \text{ bel (B)},$$

named in honor of Alexander Graham Bell, scientist, inventor, engineer and innovator credited with patenting the first practical telephone.

1 B = 10dB "10 decibels"





The Decibel for Amplitude Ratios

Given two signals S_1 and S_2 , whose magnitudes $|S_1|$ and $|S_2|$ are such that: $|S_2| = 10|S_1|$, the power ratio is given by:

$$\frac{P_2}{P_1} = \frac{|S_2|^2}{|S_1|^2} = 100, \text{ which corresponds to } \log_{10}(100) = 2B = 20\text{dB}$$

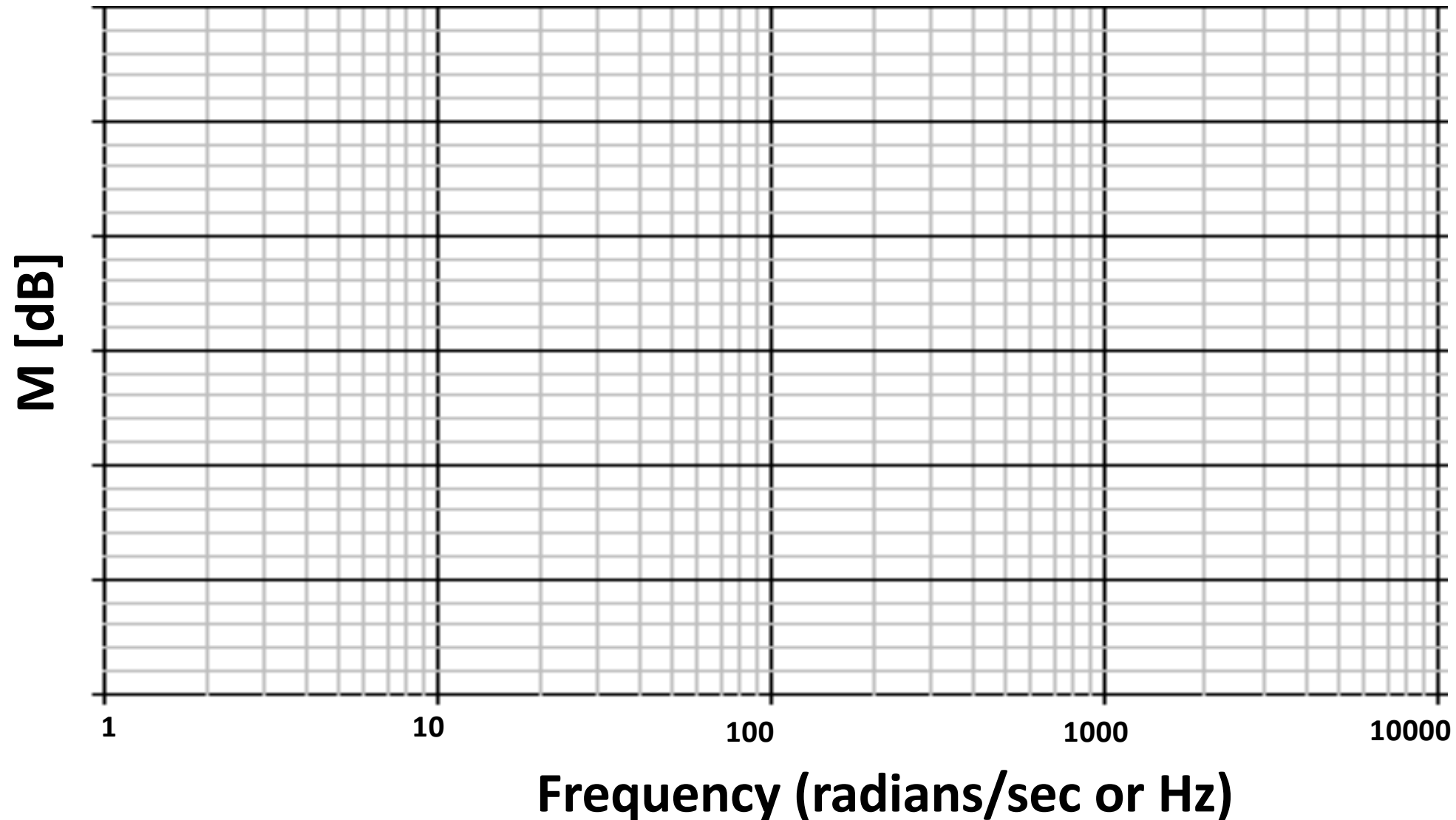
Therefore, for a transfer function magnitude $|\hat{\mathbf{H}}(\omega)| = M$, we can express M in decibel units as follows:

$$M \text{ [dB]} = 20 \log M = 20 \log |\hat{\mathbf{H}}(\omega)|$$



The Bode Magnitude Plot

Plots the magnitude M expressed in dB, with frequency on a logarithmic axis.

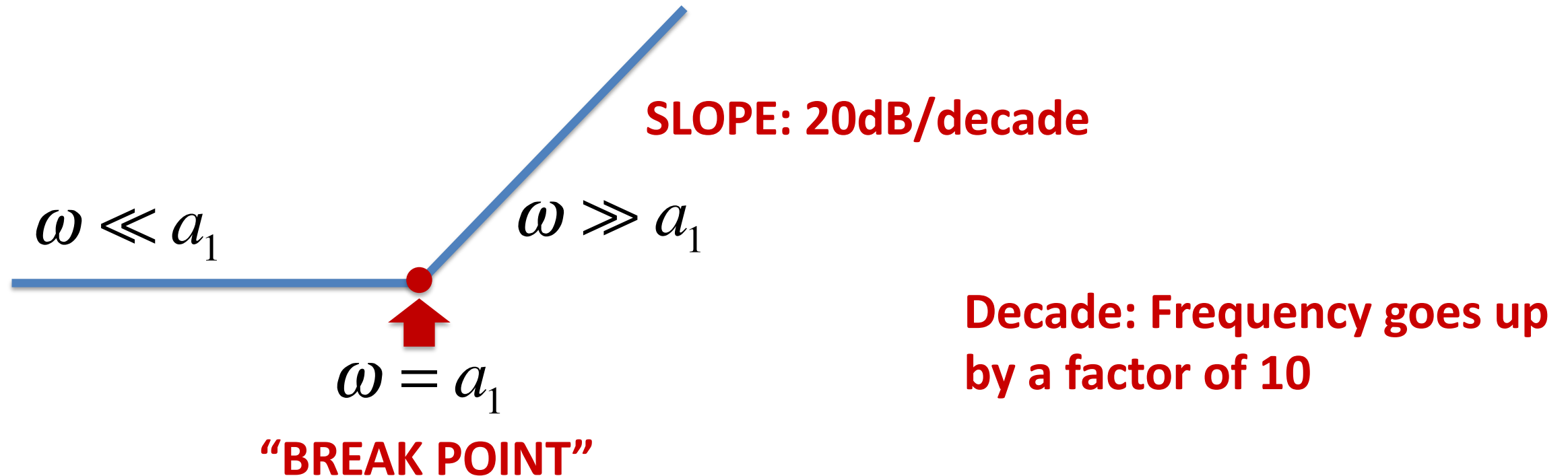




Plotting Zeros

$$20\log\left|1 + \frac{j\omega}{a_1}\right| = 20\log\sqrt{1 + \left(\frac{\omega}{a_1}\right)^2} \approx \begin{cases} 20\log(1) = 0 & \text{when } \omega \ll a_1 \\ 20\log\left(\frac{\omega}{a_1}\right) & \text{when } \omega \gg a_1 \end{cases}$$

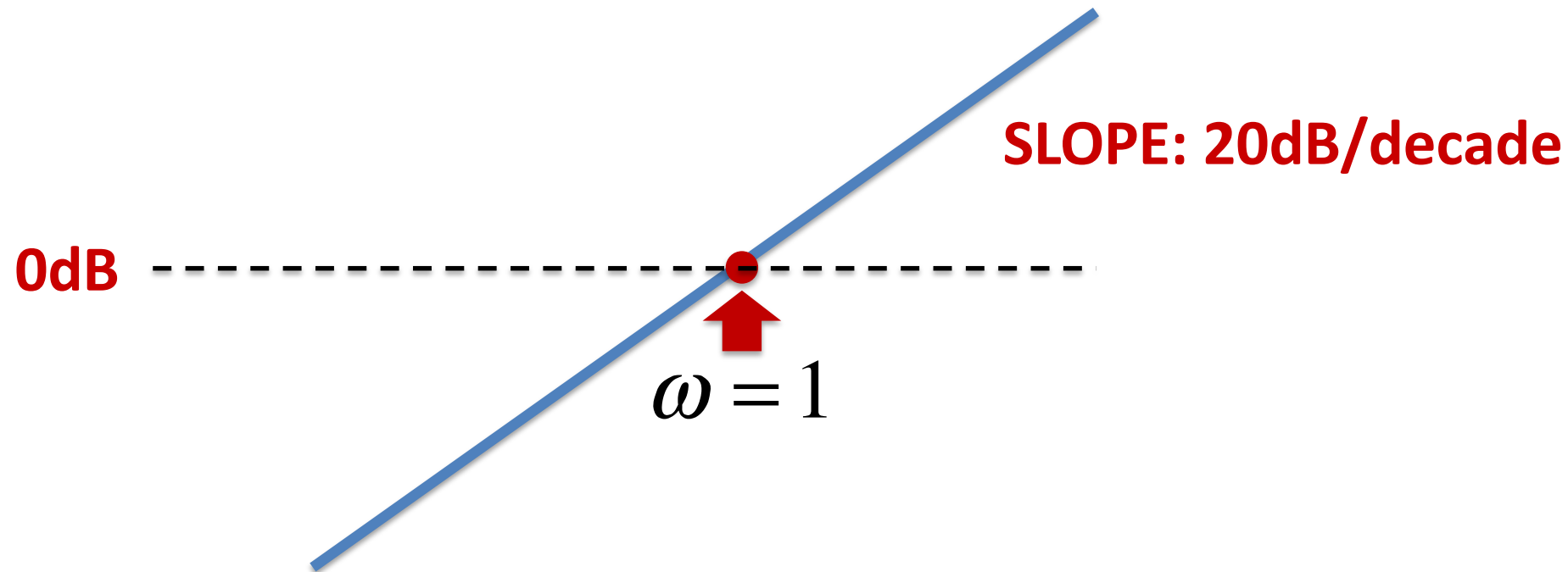
APPROXIMATED BY A “RAMP” PATTERN ON A LOGARITHMIC PLOT





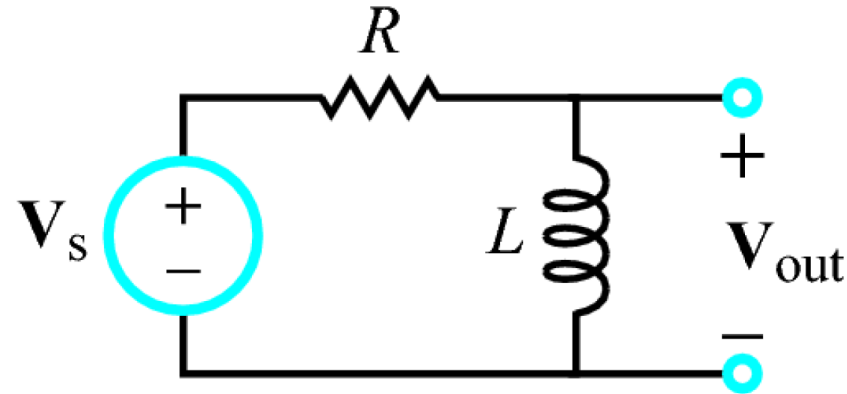
Special Case: A Zero at 0

$$20 \log \left| \frac{j\omega}{\omega_c} \right| = 20 \log \left(\frac{\omega}{\omega_c} \right) \quad \text{STRAIGHT LINE ON A LOGARITHMIC PLOT}$$





RL Circuit Example



$$\hat{\mathbf{H}}(\omega) = \frac{\mathbf{V}_{out}}{\mathbf{V}_s} = \frac{j\omega L}{R + j\omega L} = \frac{j(\omega / \omega_c)}{1 + j(\omega / \omega_c)}, \text{ where: } \omega_c = R / L$$

$$M = |\hat{\mathbf{H}}(\omega)| = \frac{(\omega / \omega_c)}{\sqrt{1 + (\omega / \omega_c)^2}}$$

$$M [\text{dB}] = 20 \log M = 20 \log(\omega / \omega_c) - 20 \log \sqrt{1 + (\omega / \omega_c)^2}$$

1

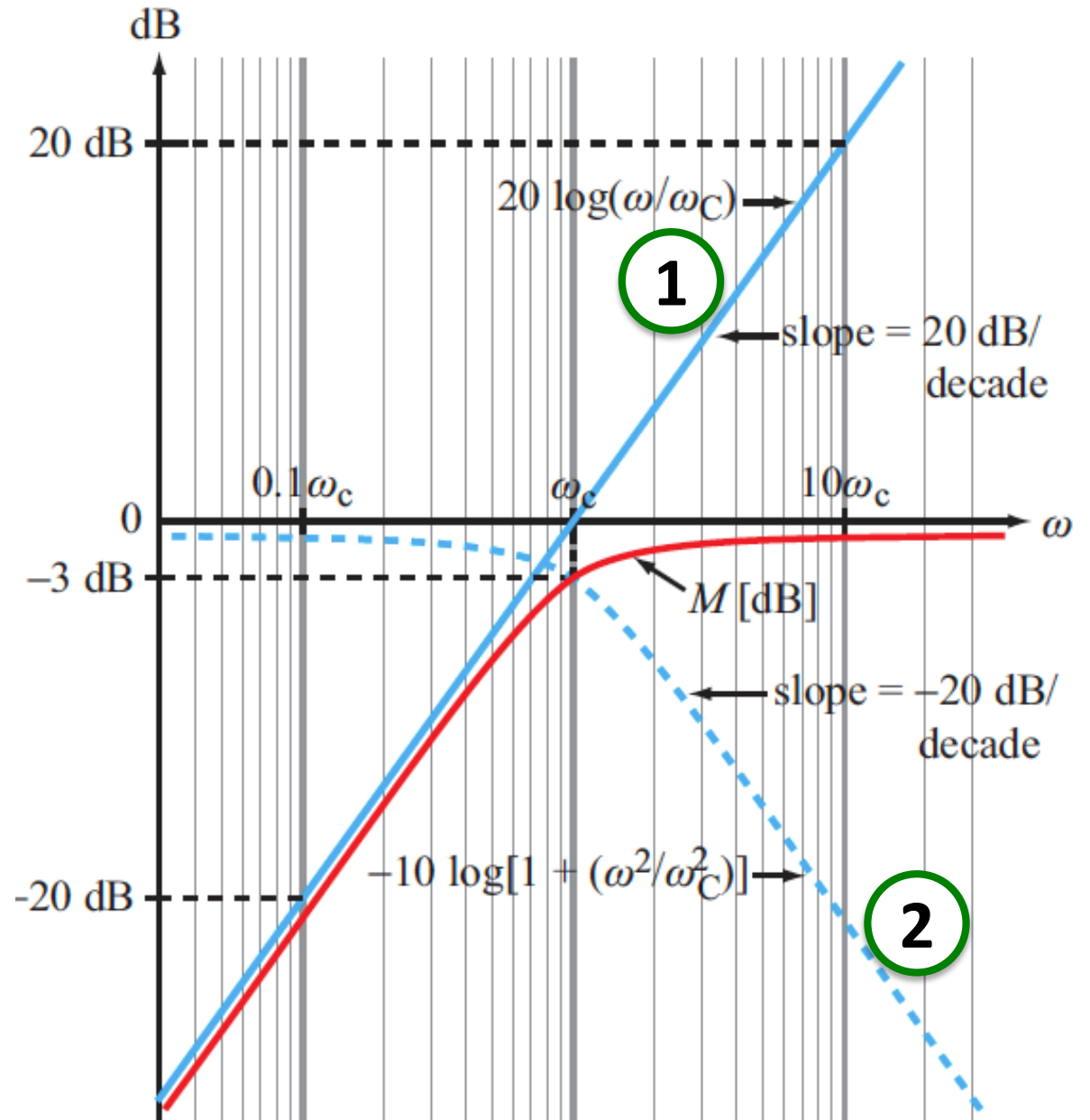
2



The Bode Magnitude Plot

$$M [\text{dB}] = 20 \log(\omega / \omega_c) \quad \textcircled{1}$$
$$-10 \log(1 + (\omega / \omega_c)^2) \quad \textcircled{2}$$

On the frequency axis,
every tick is called a
“decade” since it
represents a factor of 10





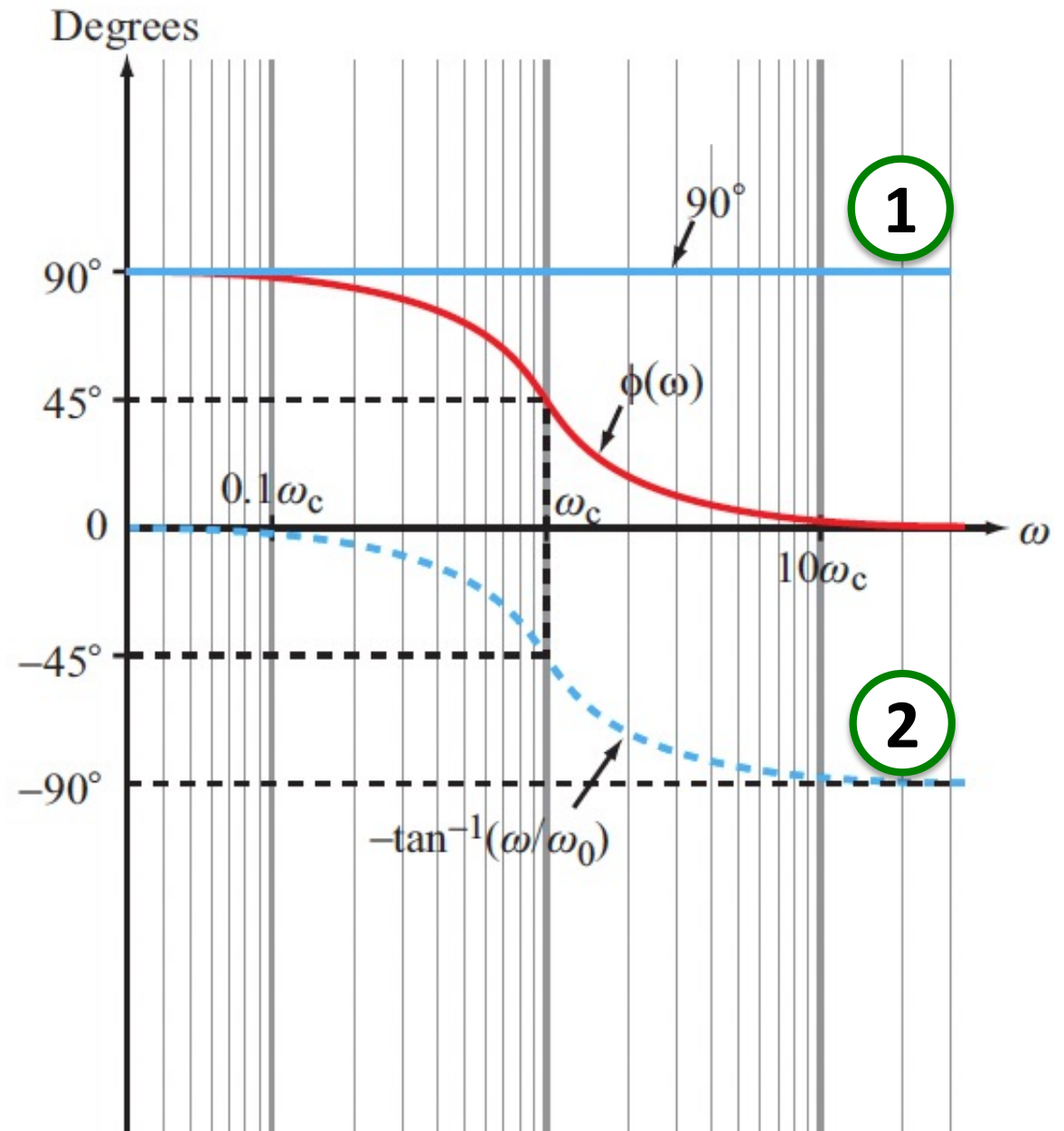
The Bode Phase Plot

$$\hat{H}(\omega) = \frac{j(\omega / \omega_c)}{1 + j(\omega / \omega_c)}$$

$$\begin{aligned}\phi(\omega) &= \phi_{\text{numerator}} - \phi_{\text{denominator}} \\ &= 90^\circ - \tan^{-1}(\omega / \omega_c)\end{aligned}$$

①

②





Bode Plot Example #2

Sketch the Bode Magnitude Plot for the following system:

$$\mathbf{H}(\omega) = \frac{(20 + j5\omega)(20 + j\omega)}{j\omega}$$

Step 1: Re-write the system in normalized form:

$$\mathbf{H}(\omega) = \frac{20\left(1 + \frac{j5\omega}{20}\right) \times 20\left(1 + \frac{j\omega}{20}\right)}{j\omega} = 400 \frac{\left(1 + \frac{j\omega}{4}\right)\left(1 + \frac{j\omega}{20}\right)}{j\omega}$$

One pole at $\omega_c = 0$

Two zeros at $\omega_c = 4, 20$ rad/sec

Constant gain of 400 ($20 \log_{10}(400) = 52\text{dB}$)



Breakpoints

***Breakpoints* are values of ω where there are changes in slope on the magnitude or phase Bode plot.**

- We find them by setting the imaginary part of the terms $(a + j\omega b)$ or $(1 + j\omega/a)$ equal to the real part.

$$\mathbf{H}(\omega) = 400 \frac{\left(1 + \frac{j\omega}{4}\right)\left(1 + \frac{j\omega}{20}\right)}{j\omega}$$

- In the normalized form (above), breakpoints occur at $\omega = a$ in the terms of the form $(1 + j\omega/a)$.
- For the term $j\omega$, the breakpoint is at $\omega = 0$ (since real part is 0).
- Zeros are breakpoints in the numerator: (4, 20) for our example.
- Poles are breakpoints in the denominator: (0) for our example.

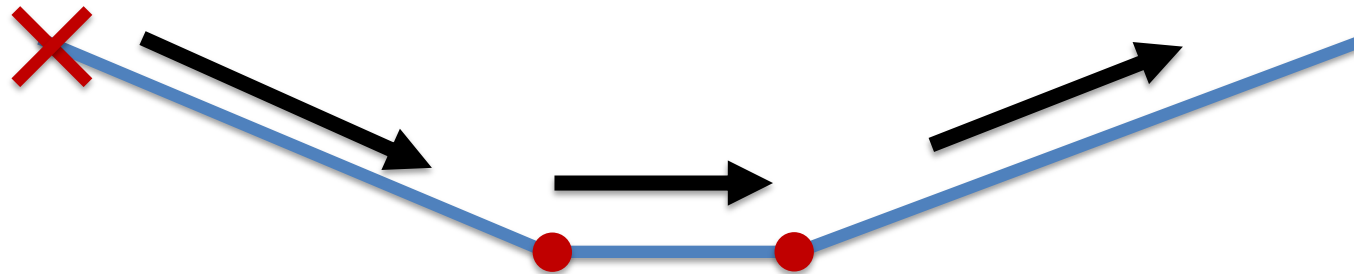


Bode Plot Example #2

The formula for Magnitude in dB:

$$M[dB] = 20 \log |400| + 20 \log \left| 1 + \frac{j\omega}{4} \right| + 20 \log \left| 1 + \frac{j\omega}{20} \right| - 20 \log |\omega|$$

pole at 0 causes
-20 dB/decade slope



zero at 4 rad/s
causes 20
dB/decade
increase in slope

zero at 20 rad/s
causes 20
dB/decade
increase in slope



The Magnitude Plot

To fix the plot vertically, we need the gain at one convenient value of ω ...

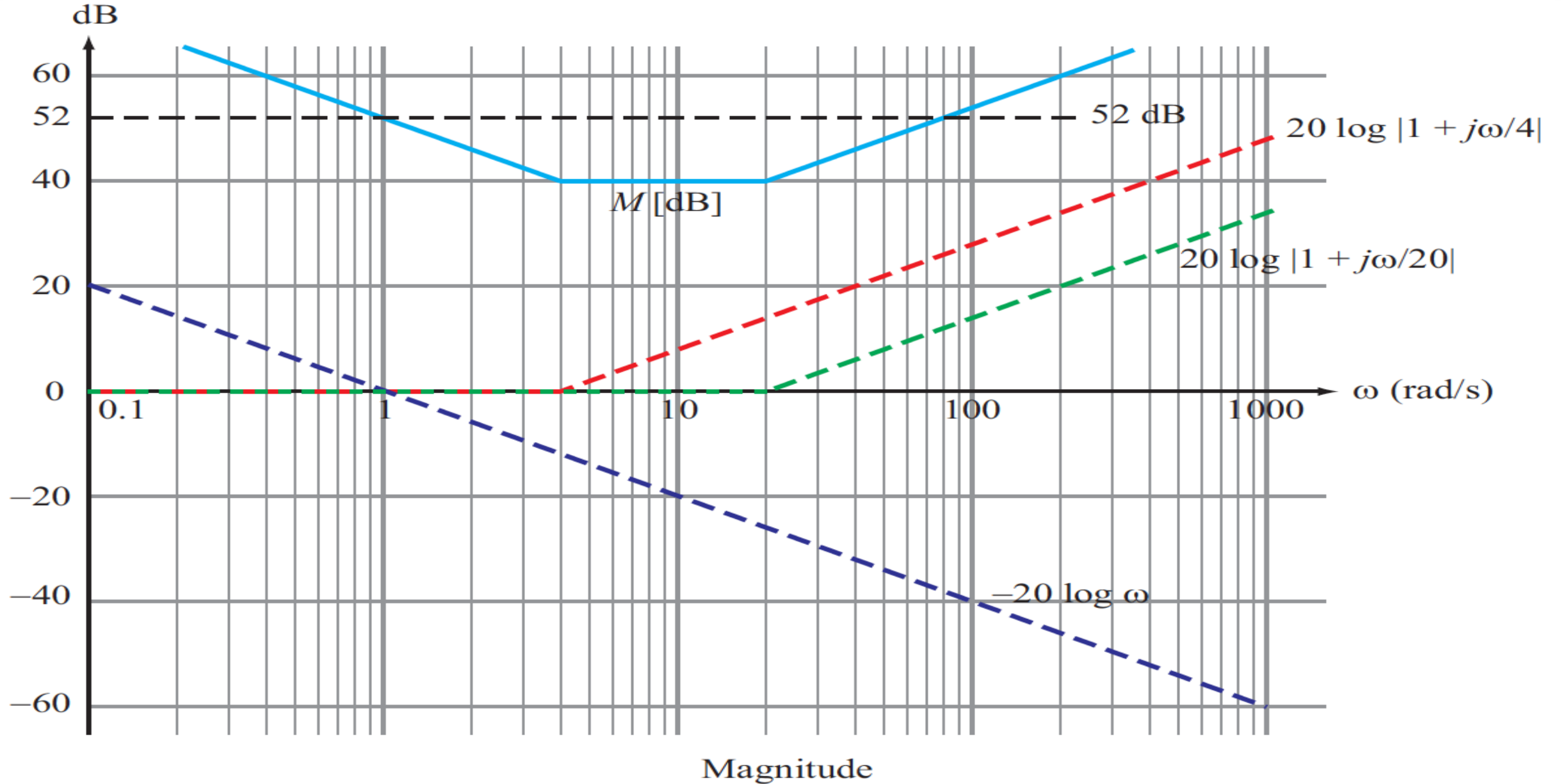
$$\hat{\mathbf{H}}(\omega) = \frac{(20 + j5\omega)(20 + j\omega)}{j\omega}$$

In the range $\omega = 4$ rad/sec to $\omega = 20$ rad/sec, we will have a constant gain, which we can evaluate somewhere convenient in the middle:

$$\left| \hat{\mathbf{H}}(\omega = 10) \right| = \left| \frac{(20 + j5(10))(20 + j(10))}{j(10)} \right| = 120$$
$$\Rightarrow 20 \log_{10}(120) = 40dB$$

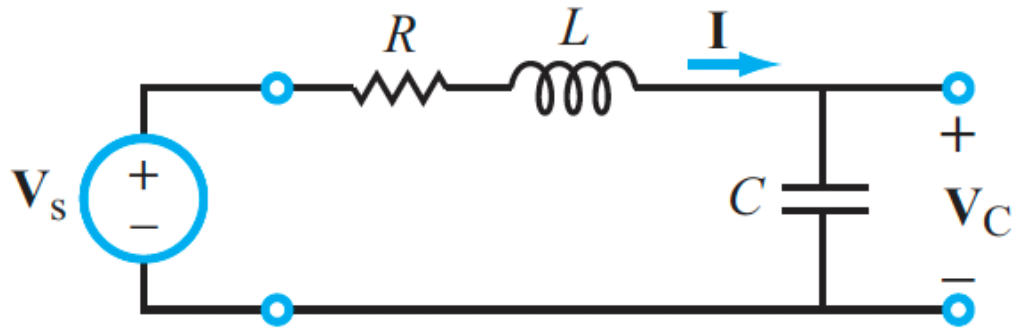


Magnitude Plot for Example #2





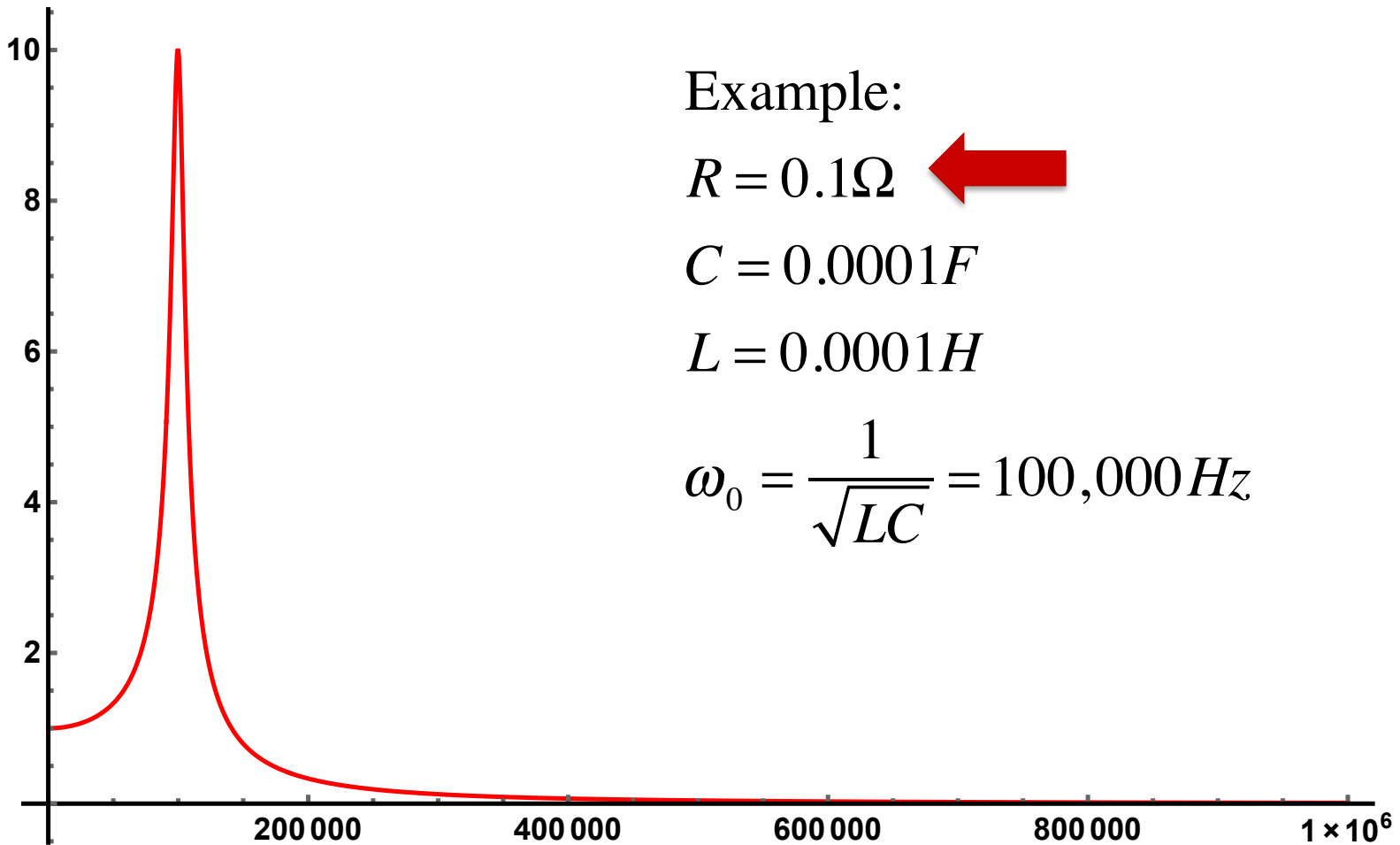
Recap: RLC Filter



$$|\hat{\mathbf{H}}_C(\omega)| = \frac{1}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$

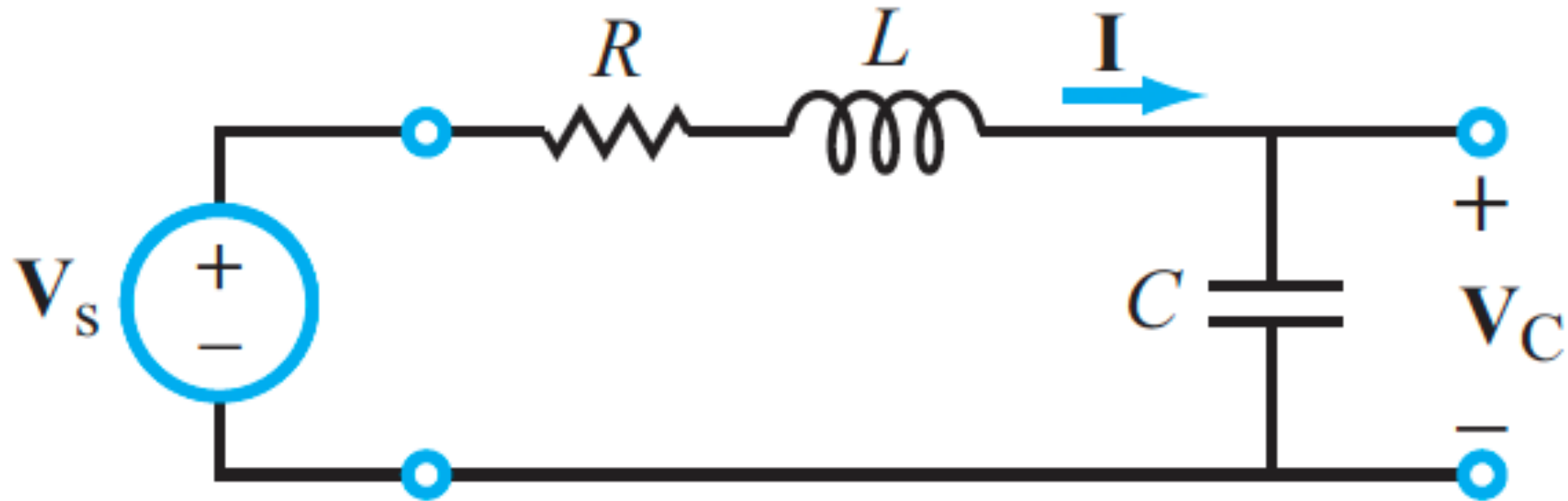
when $\omega^2 LC = 1$,

$$\omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow |\mathbf{H}_C(\omega)| \rightarrow \frac{1}{R} \sqrt{\frac{L}{C}}$$





This is a 2nd order Filter

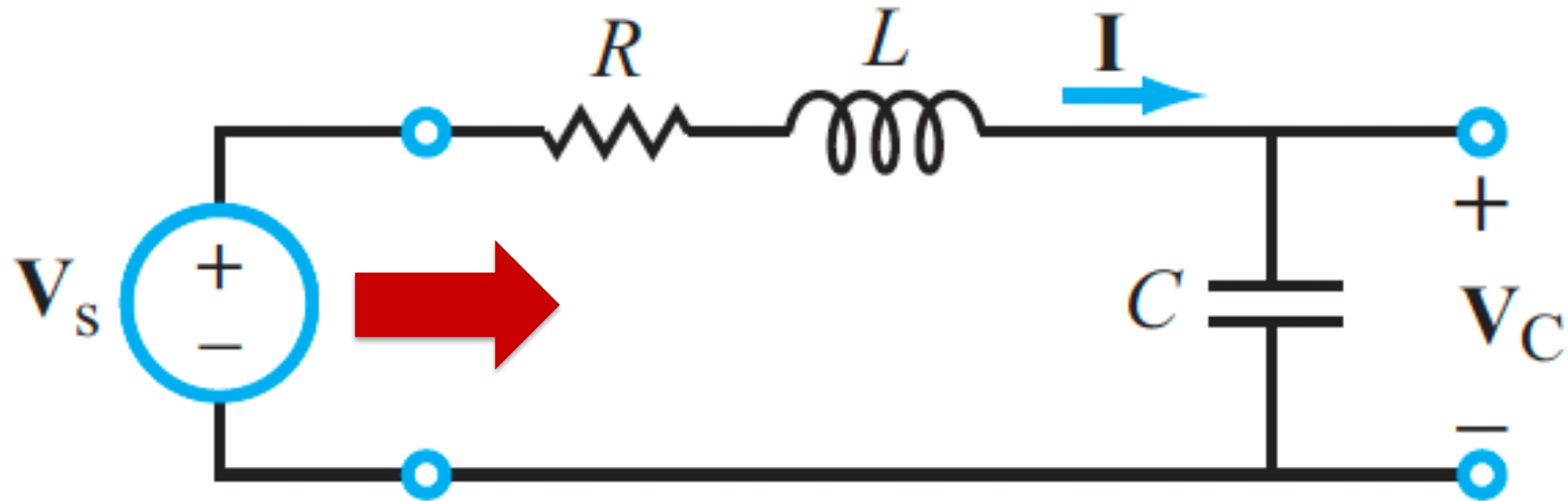


$$|\hat{\mathbf{H}}_C(\omega)| = \frac{1}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$

Order of the denominator is 2



Capacitor Voltage V_C



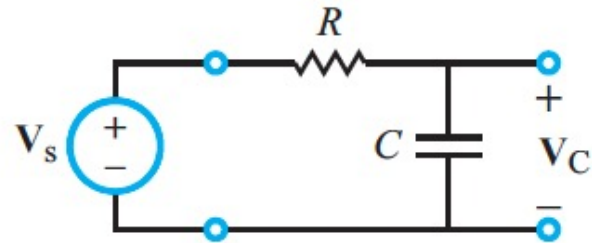
At resonance, the input impedance is purely resistive

$$\mathbf{Z}_L + \mathbf{Z}_C = j\omega L + \frac{1}{j\omega C} = j\left(\omega L - \frac{1}{\omega C}\right)$$

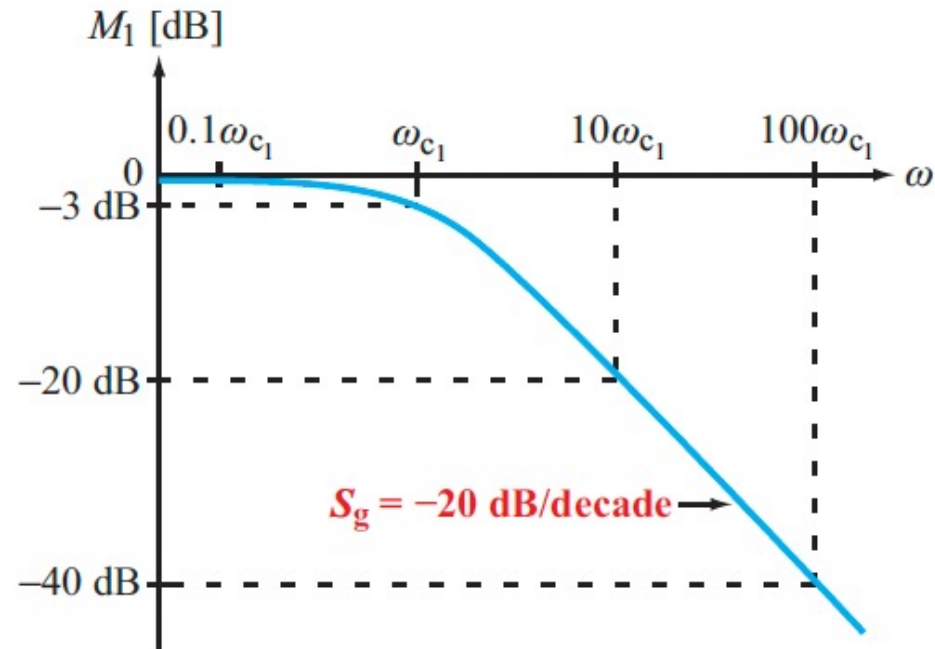
This is 0 when $\omega L = \frac{1}{\omega C}$ or $\omega = \frac{1}{\sqrt{LC}}$



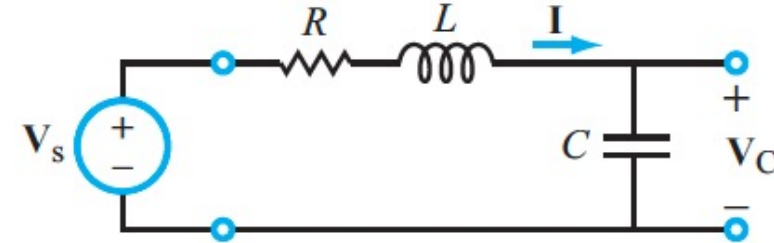
Higher-order Filter Has Steeper Roll-off



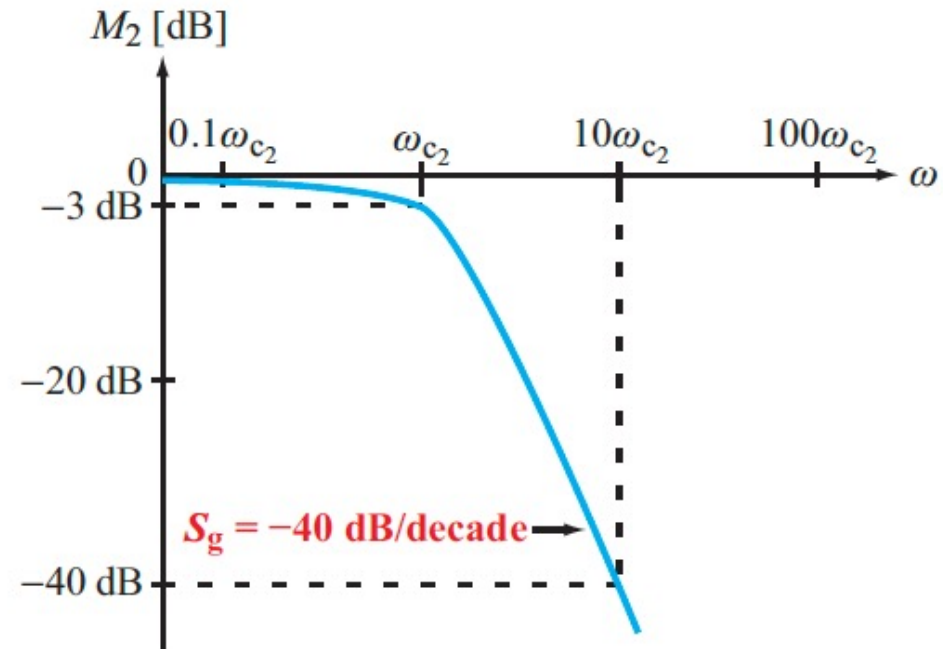
(a) First-order filter



(b) Response of first-order filter



(c) Second-order filter



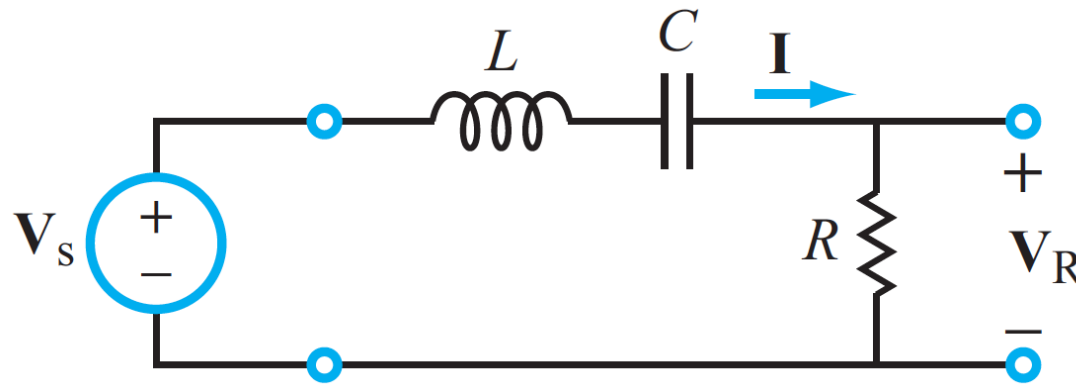
(d) Response of second-order filter

Raising the order of the filter (adding higher-order poles) is one possible strategy since higher-order filters provide steeper roll-offs (20dB/decade per pole)



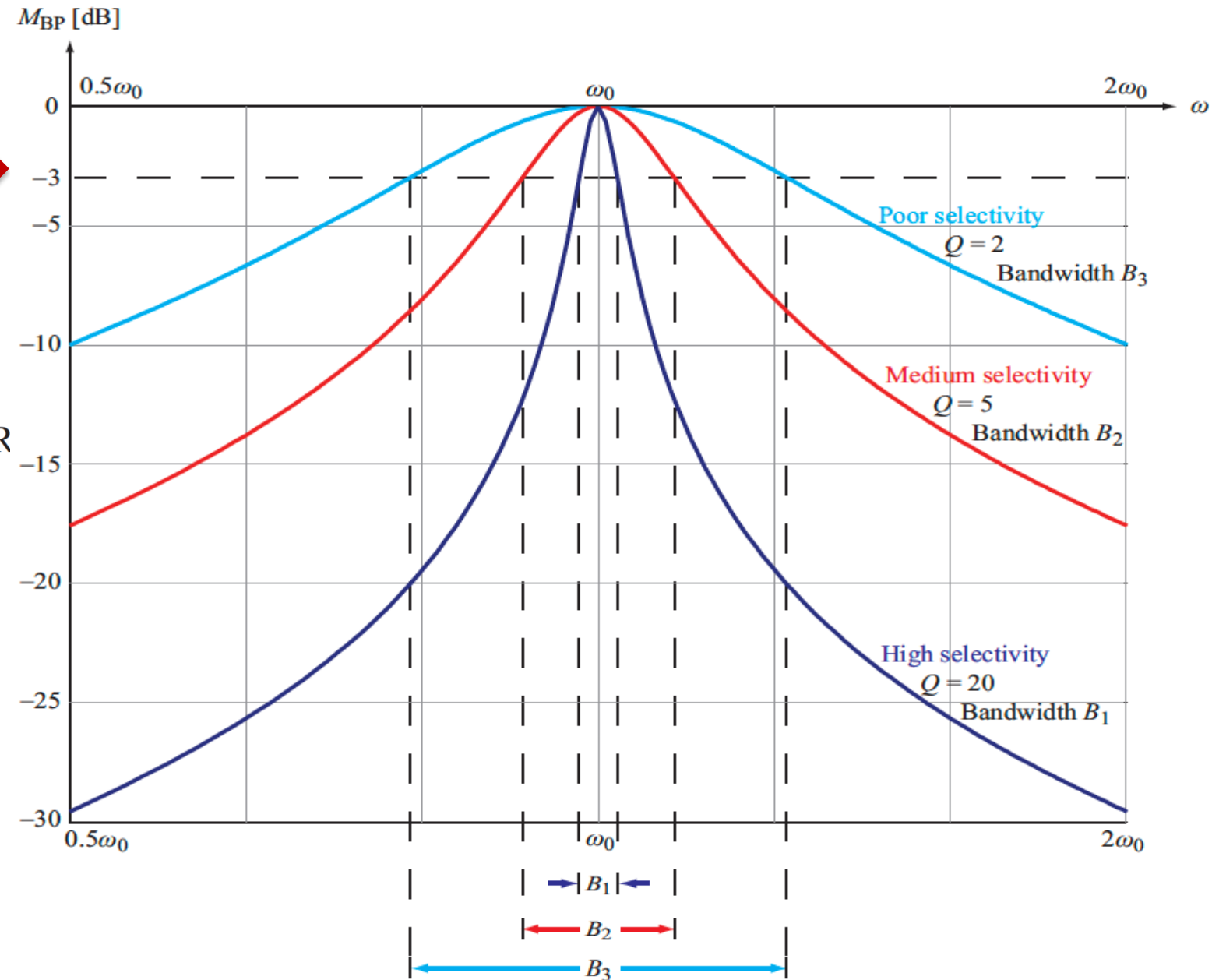
Recap: Filter Quality Factor Q

$$Q = \frac{\omega_0}{B} = \frac{\omega_0 L}{R}$$



Q is a dimensionless ratio used to measure the selectivity of a filter

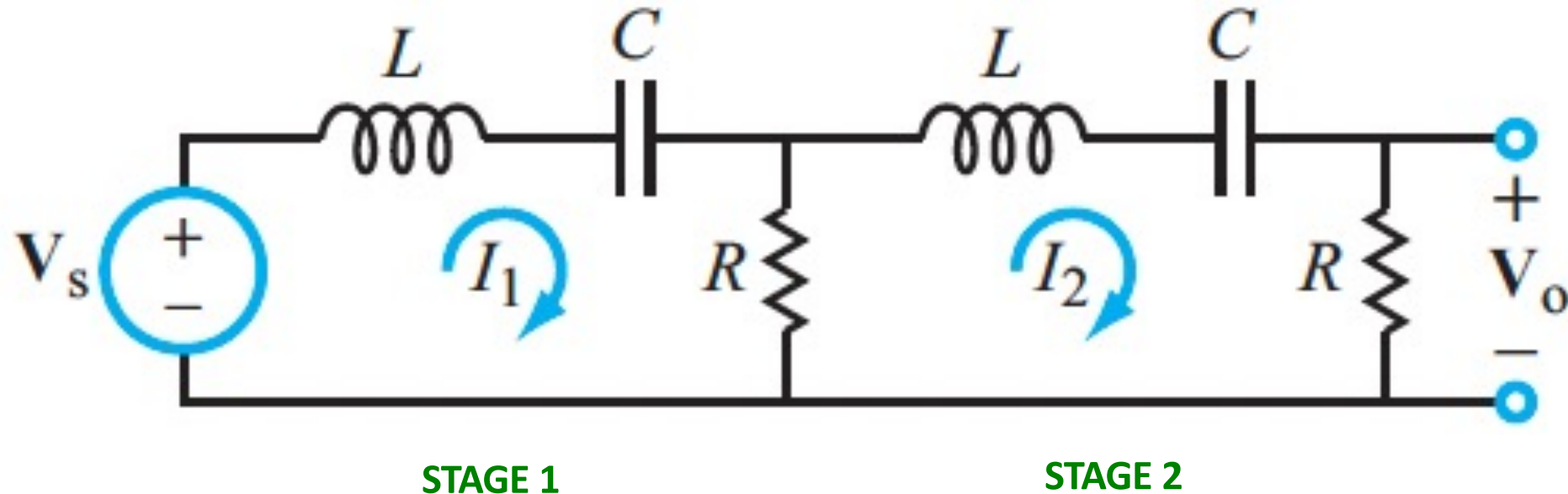
Lower R values yield higher Q filters





The Next Step: A 2-Stage Filter

We can build higher-order filters by cascading simpler filters, for example:



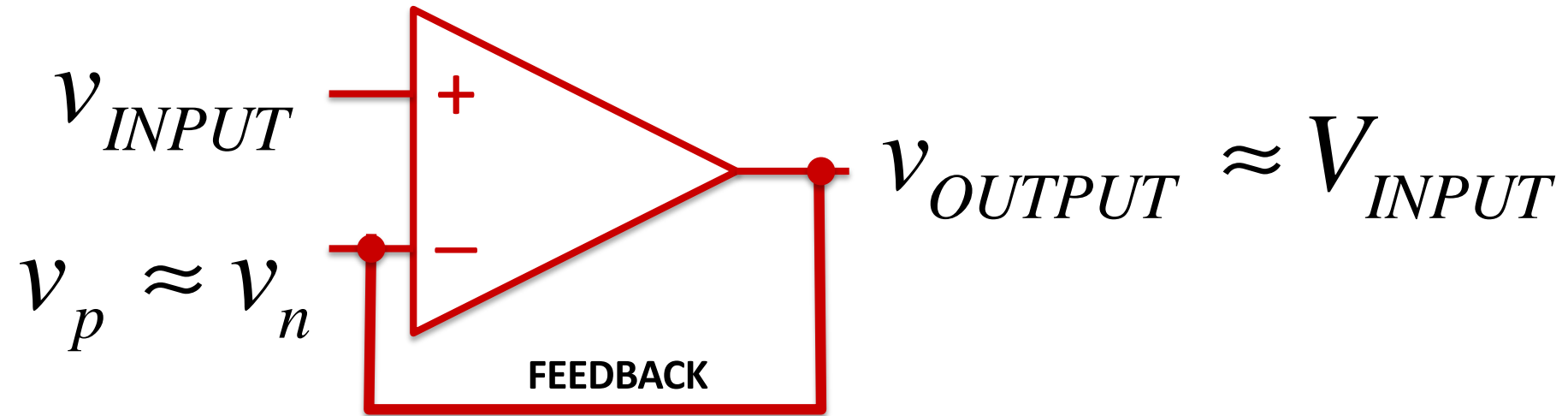
Messy to analyze, and hard to design!

$$\hat{H}(\omega) = \frac{\omega^2 R^2 C^2}{\omega^2 R^2 C^2 - (1 - \omega^2 LC)^2 - j3\omega RC(1 - \omega^2 LC)}$$

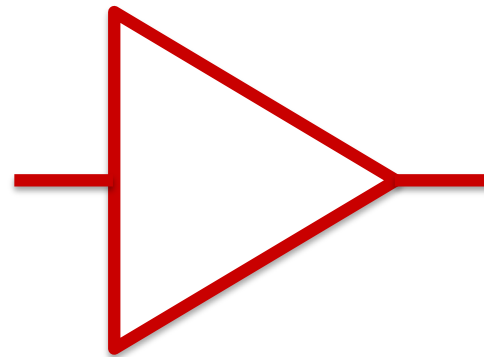
THIS IS NOT PRACTICAL FOR SCALING UP



Recap: Voltage Follower



SIMPLIFIED SYMBOL: BUFFER

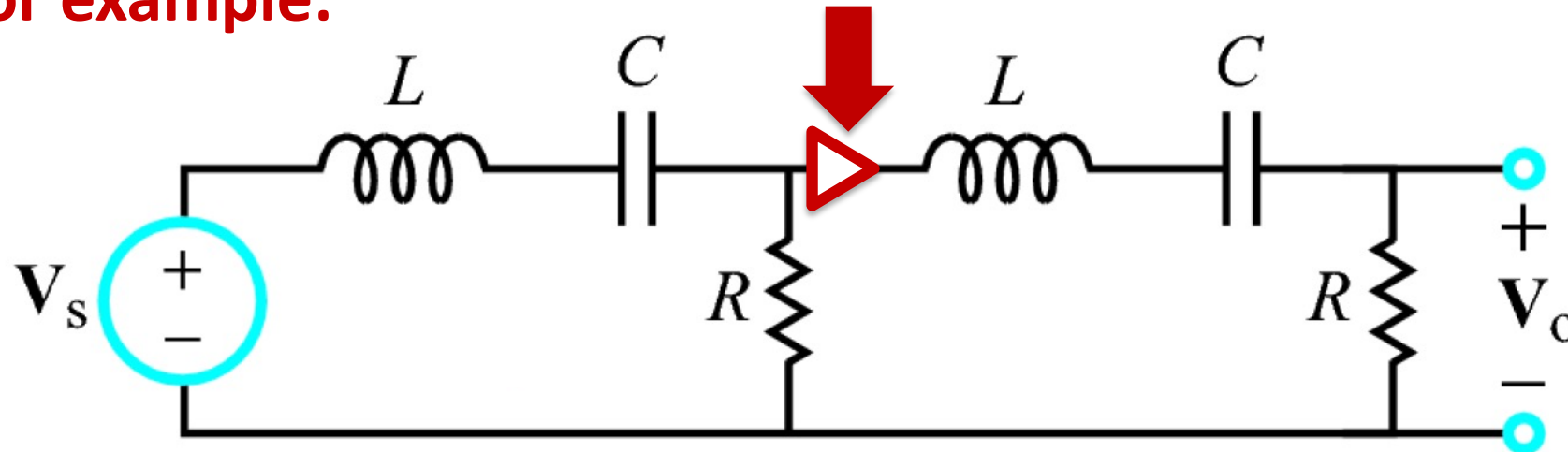


ADDING A VOLTAGE FOLLOWER BETWEEN STAGES IS ONE WAY TO SIMPLIFY



Buffered 2-Stage Filter

We can build more effective filters by cascading simpler filters with buffering, for example:



- **Much easier to analyze and design!**
 - Each stage contributes an additional product term to $H(\omega)$

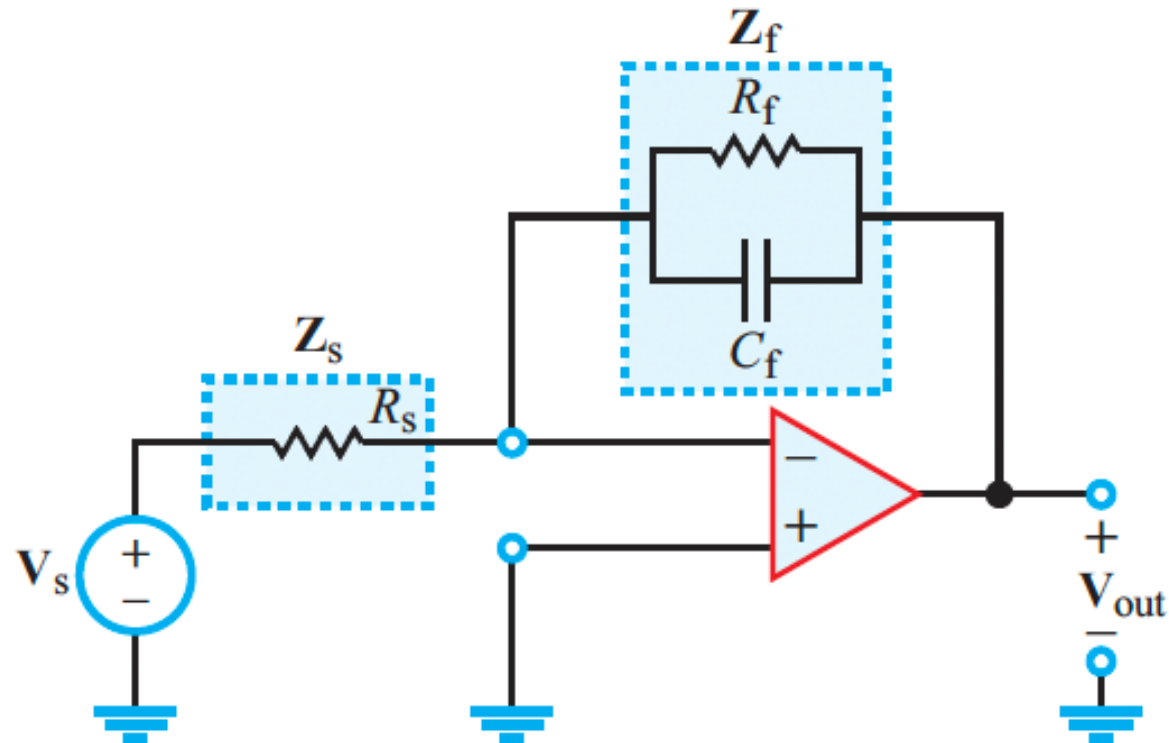
$$\hat{\mathbf{H}}(\omega) = \hat{\mathbf{H}}_R(\omega)\hat{\mathbf{H}}_R(\omega)$$

GENERALLY, OP-AMPS ARE A FILTER DESIGNER'S BEST FRIEND



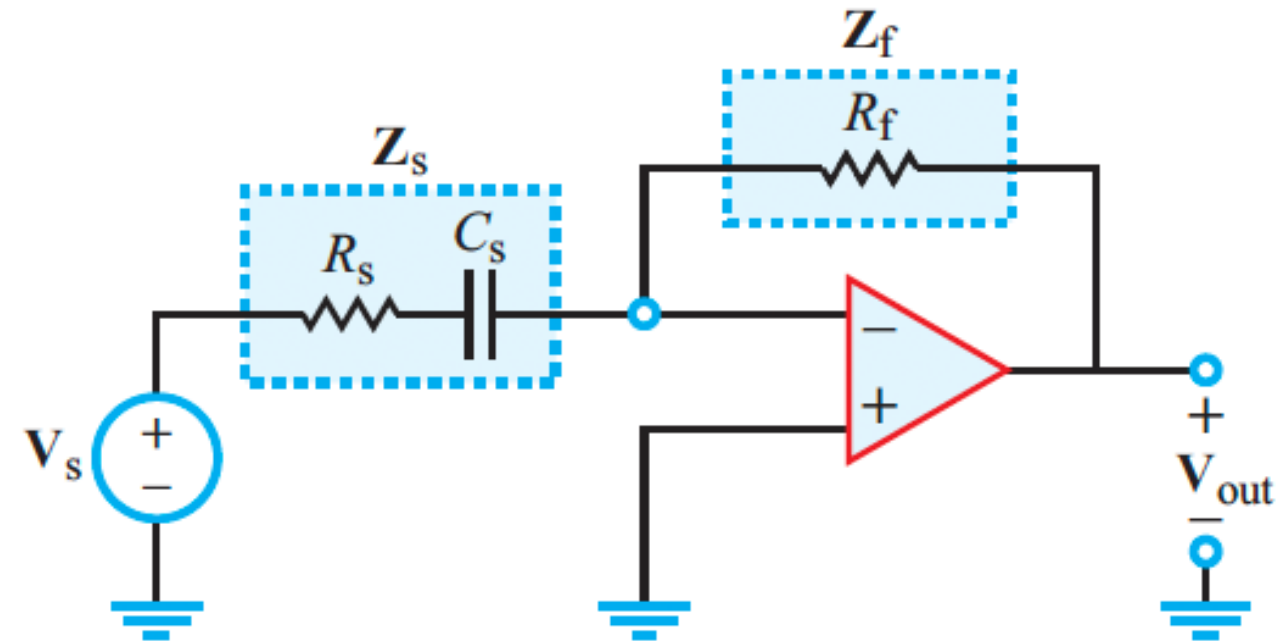
Recap: Op-AMP Filters

ACTIVE LOW-PASS FILTER



$$\hat{\mathbf{H}}_{LP}(\omega) = -\frac{R_f}{R_s} \left(\frac{1}{1 + j\omega R_f C_f} \right)$$

ACTIVE HIGH-PASS FILTER

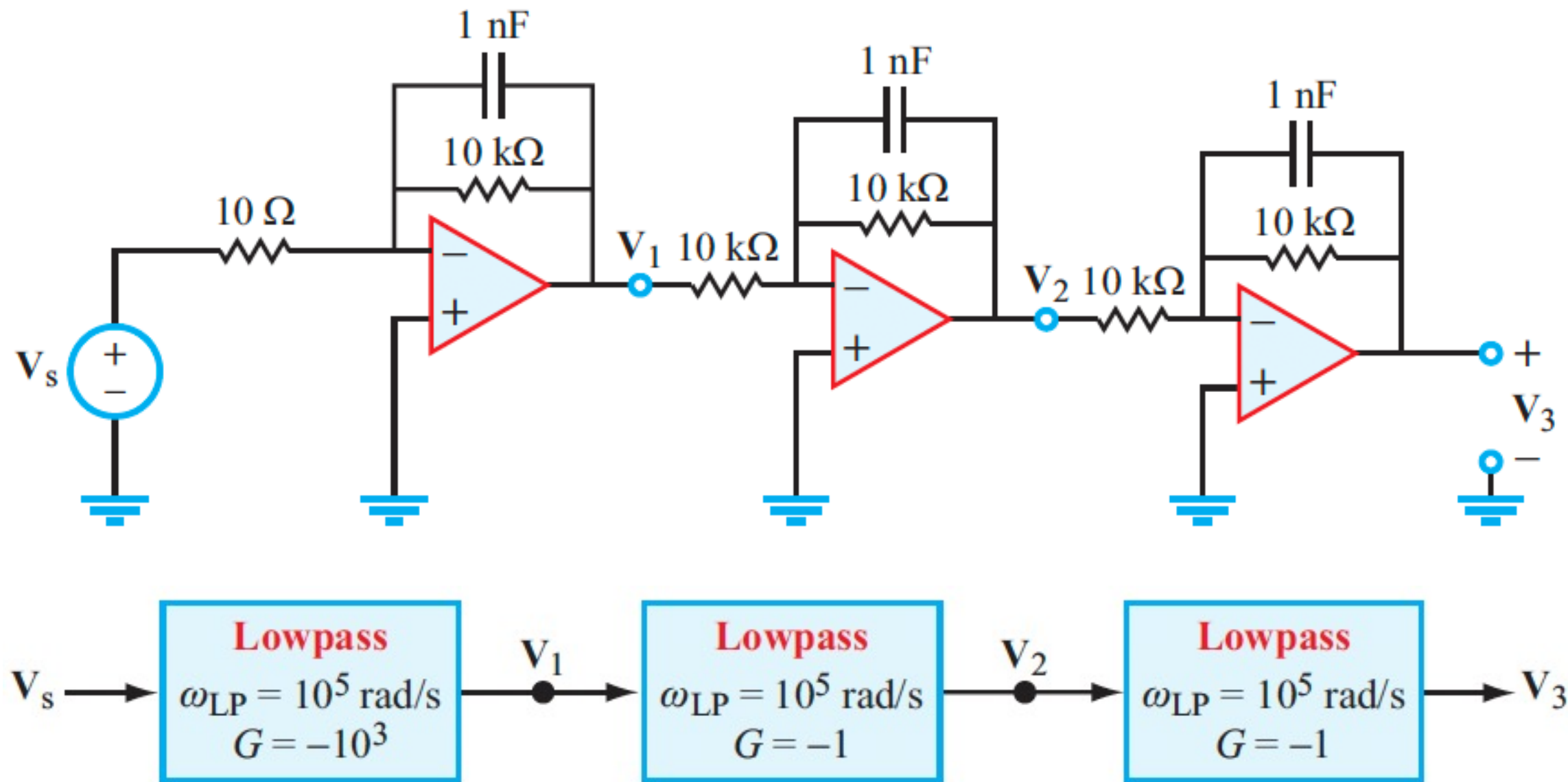


$$\hat{\mathbf{H}}_{HP}(\omega) = -\frac{R_f}{R_s} \left(\frac{j\omega R_f C_f}{1 + j\omega R_f C_f} \right)$$

WE CAN ACHIEVE THE DESIRED GAIN VALUE AND POLE LOCATION BY CHOOSING COMPONENTS

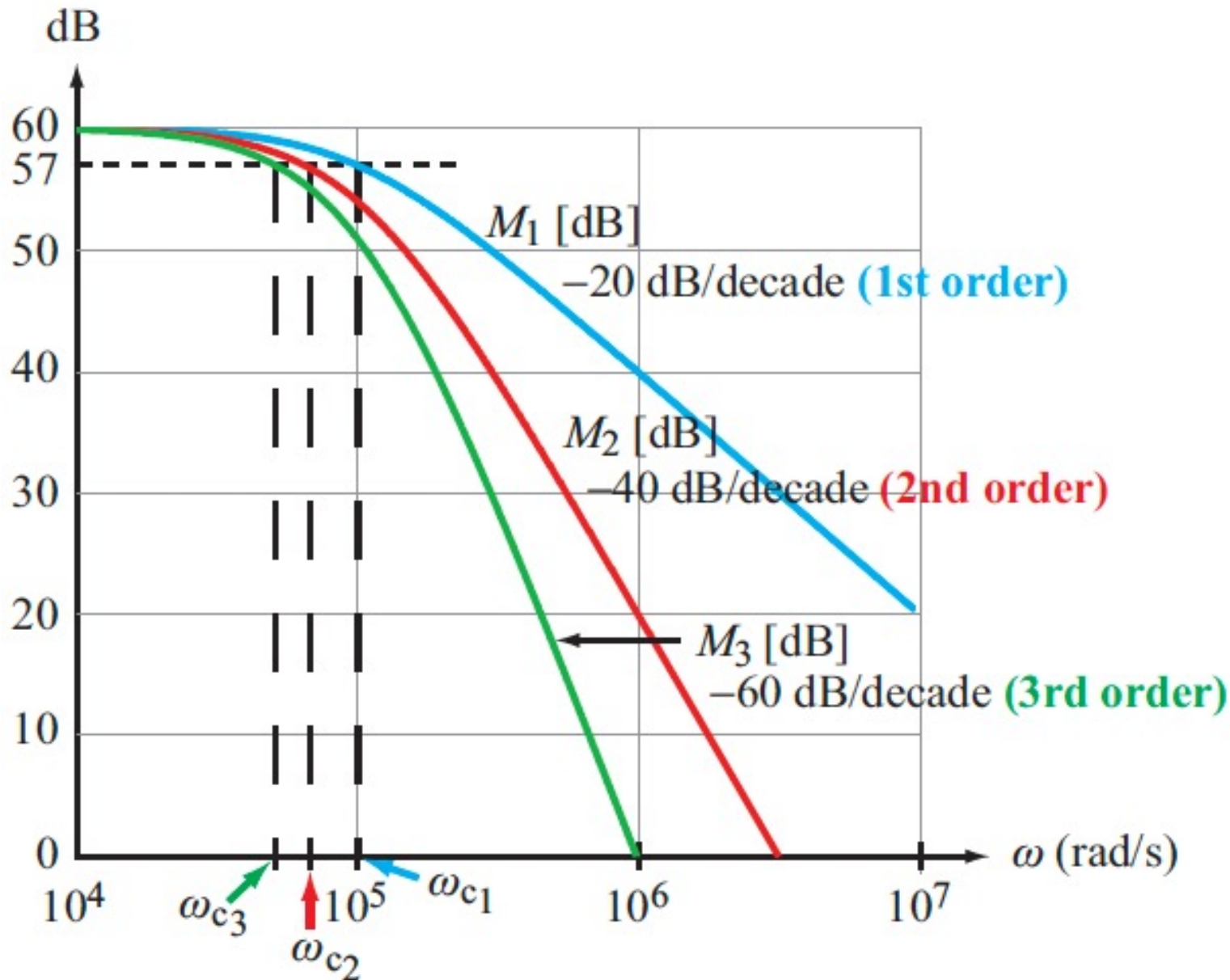


Example: 3-Stage Low-pass Filter





Example: 3-Stage Low-pass Filter



MINOR, BUT SOLVABLE ISSUE:

The half-power corner frequencies are affected by the cascading:

Stage 1: $\omega_{c1} = 10^5 \text{ rad/sec}$

Stage 2: $\omega_{c2} = 6.4 \times 10^4 \text{ rad/sec}$

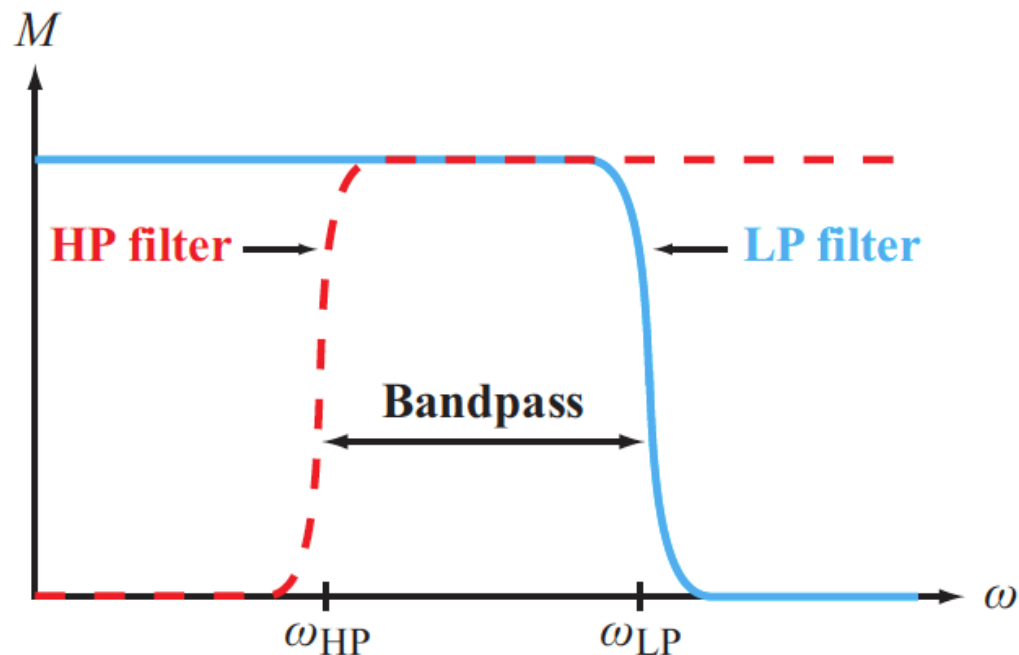
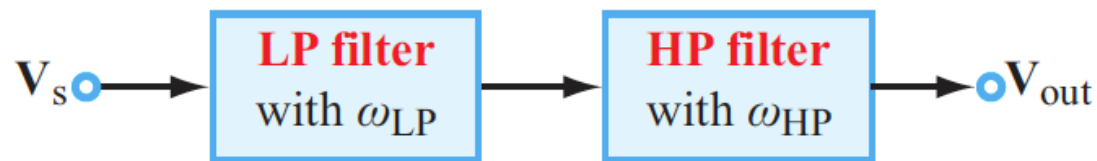
Stage 3: $\omega_{c3} = 5.1 \times 10^4 \text{ rad/sec}$



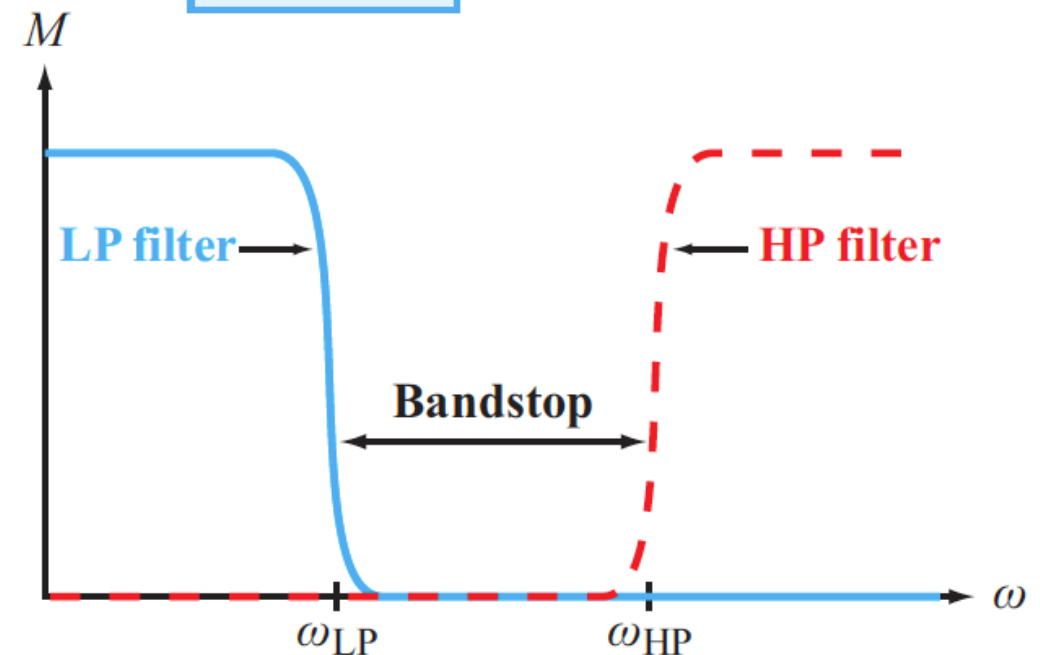
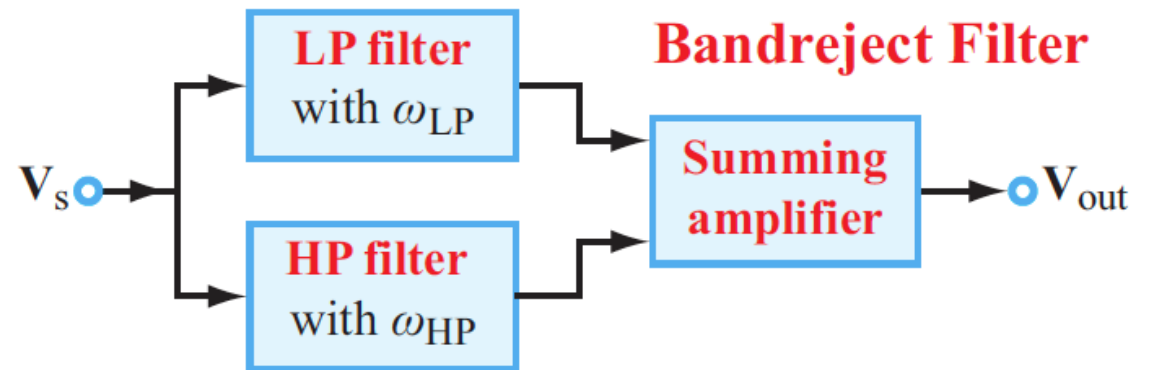
Combining Filters

Basic filters can be combined in series and/or parallel to build more complex or customized filters

Bandpass Filter



Bandreject Filter





Summary

- **We looked at passive and active filters today**
 - **Passive filters** are basically RLC circuits / LTI systems that we are already familiar with.
 - However, we looked at them from the fresh perspective of signal filtering in a frequency-dependent manner
 - **Active filters** bring amplifiers into the picture, and add new capabilities to passive filters
 - They make it easy to combine building basic blocks to build more sophisticated filters in a modular fashion
 - They can provide a gain that exceeds what a passive filter can provide
 - We did not discuss it, but active filters can be built without inductors (clumsy, bulky, hard to miniaturize!)



Next Class

- **Work through the textbook examples in Section 6-13**
- **We will derive the sampling theorem, and study the practical methods needed to apply it**
- **The Final exam is coming up**
 - **It is a shorter format compared to previous semesters**
 - **Plenty of past examples are posted on the Canvas system**
 - **We are happy to review any topic and hold study sessions**
 - **Please let us know what topics or examples you need!**